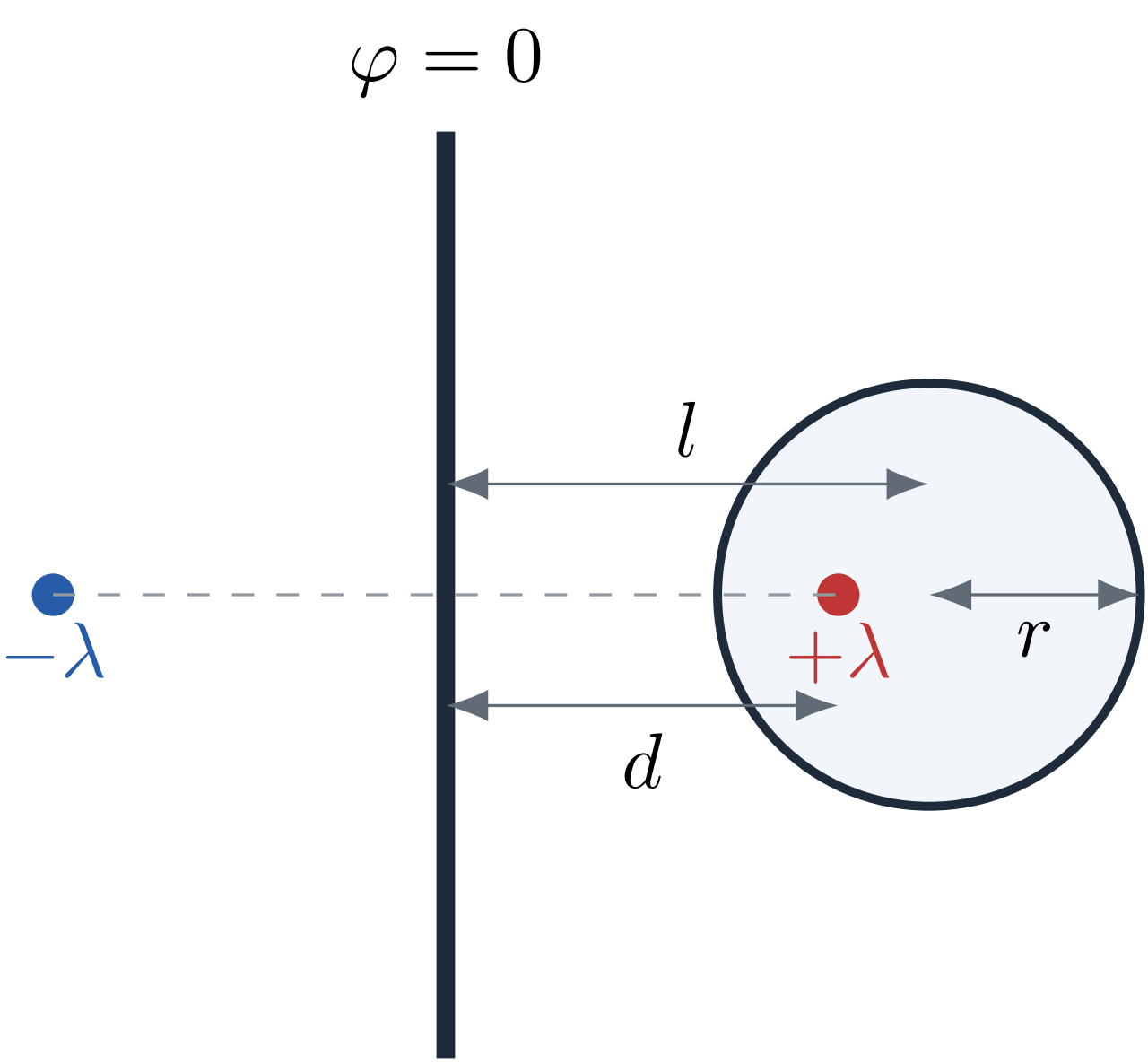

$$d^2 = l^2 - r^2$$

$$C' = \frac{2\pi\varepsilon}{\operatorname{arccosh}(l/r)}$$